

PROTIK DEV

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Summary

Full-stack software engineer (TypeScript, Next.js, Go) with ~3 years of corporate experience and 2+ years of freelance experience, building products from UI to production infrastructure.

Education

BGC Trust University

B.Sc. in Computer Science and Engineering

Chattogram, Bangladesh

Professional Experience

Devspace

Mar 2026 – Aug 2026

Software Engineer

- Built full-stack React and TypeScript features on an **agentic AI pipeline** (Mastra AI, Convex, Vercel AI SDK).
- Built a **dynamic tool registry**: agent tools are stored in the database and discovered at runtime, not hardcoded.
- Contributed to CI/CD pipeline configuration and deployment automation.
- Tech:** *React, TypeScript, Mastra AI, Convex, Vercel AI SDK, GitHub Actions, Codex*

Osilion

Jul 2025 – Feb 2026

Full-Stack Engineer

- Served as one of the **primary engineers** on a client's internal tool for improving employee productivity.
- Worked on an intelligent recruitment platform, contributing to its technical architecture, database design, and component design patterns.
- Built SSR, ISR, and SSG features with SEO metadata and Open Graph tags.
- Mentored a junior developer through regular code reviews.
- Tech:** *Next.js, TypeScript, Tailwind, shadcn/ui, Zustand, Appfarm (low-code), Azure*

Hone

Aug 2025 – Sep 2025

Frontend Engineer

- Implemented a **Smart Scaling system** that automatically adapts the desktop app to smaller screens.
- Owned the review and download pages end to end; found and fixed a small-screen layout issue with a CSS grid pattern, validated through manual cross-device testing.
- Improved code splitting and lazy loading to cut initial load weight, and fixed a keyboard focus-visibility issue.
- Tech:** *React, TypeScript, Tailwind, shadcn/ui*

Hello World Communications

Sep 2024 – Jul 2025

Full-Stack Engineer

- Led the development flow end to end** across multiple concurrent client projects, conducting code reviews and making key architectural decisions.
- Reduced client closing time by **roughly 50%** through better scoping and client-to-team delivery alignment.
- Built backend APIs and integrated third-party services, including the **SSLCommerz payment gateway** and WebSocket real-time features.
- Delivered SSR pages for real-time data and ISR/SSG for marketing pages, with SEO work on structured metadata, sitemaps, and Lighthouse tuning.
- Mentored a junior developer across multiple client projects.
- Tech:** *Next.js, TypeScript, Go, shadcn/ui, TanStack Query, Zustand, Drizzle ORM, Appwrite, Firebase, Docker*

Bilsida

Oct 2023 – Aug 2024

Full-Stack Engineer

- Built a Swedish car marketplace **from scratch across the full stack**; it went on to attract **two of Sweden's largest car dealers**.
- Owned backend logic, database schema design, image optimization, and SEO.
- Mentored a junior developer in building a chat system.
- Tech:** *Next.js, TypeScript, Tailwind, shadcn/ui, Appwrite, Docker, Context API*

Fiverr

Jul 2021 – Oct 2023

Freelance Full-Stack Developer

Remote

- Delivered full-stack web applications and APIs for **20+ clients**, covering frontend, backend, and deployment.

Backend & Systems Projects

Spud: Discord Bot for Team Task Coordination

[Live](#) | [Source](#)

- Built and deployed to production a Discord bot for small-team task coordination with Gemini-powered features: **duplicate-task detection** that blocks redundant work at claim time, and a **scope-drift detector** that diffs a git branch against its natural-language task description and pings the owner in-channel when work strays outside the claimed scope.
- Engineered a **GitHub App integration** end to end: short-lived installation tokens, an app-level webhook pipeline with **HMAC-SHA256 verification**, per-project rate limiting, and automatic task closure on PR merge.
- Added a role-based permission model (team lead vs. server admin) enforced in application code.
- Set up **production observability**: structured JSON logging shipped to Grafana Loki, plus synthetic monitoring with uptime alerting; containerized with Docker on a hosted Turso (libSQL) database.
- Tech:** *TypeScript, Bun, discord.js, GitHub Apps API, GitHub Actions, Gemini, Vercel AI SDK, SQLite/Turso, Grafana (Loki + Synthetic Monitoring), Docker*

Result Lookup: Exam Result Publishing System

[Source](#)

- Designed and built a **high-throughput** exam result publishing system in Go, simulating Bangladesh's SSC result release with **2M results**.
- Pre-serialized results into Redis behind a **publish-gate flag for atomic release**, with cursor-based batch seeding and SCAN-based count verification before the gate flips.
- Reached a **~32k RPS peak with zero errors** across a 615k-request load test and **p99 under 105 ms at 1,000 concurrent connections**, backed by a PostgreSQL fallback with connection limiting.
- Tech:** *Go, PostgreSQL, Redis, pgx/v5, Docker, golang-migrate, slog, miniredis*

URL Health Checker: Async Job Queue

[Source](#)

- Built an async job queue with separate API and worker binaries; workers claim jobs via `SELECT ... FOR UPDATE SKIP LOCKED`, with **exponential backoff written in SQL**, atomic with status updates.
- Recovered stale jobs with a staleness threshold instead of a heartbeat, avoiding a race on bounded-duration jobs; graceful shutdown via `signal.NotifyContext`, with integration tests against real PostgreSQL.
- Tech:** *Go, PostgreSQL, pgx/v5, Docker, golang-migrate, slog*

URL Shortener API

[Live](#) | [Source](#)

- Built a **per-IP token-bucket rate limiter** on a mutex-protected generic SafeMap, with retry-on-conflict short-code generation instead of pre-checking.
- Shipped a multi-stage Docker build running as a non-root user, with **integration tests against real PostgreSQL** and the schema loaded via `//go:embed`.
- Deployed on Render with managed PostgreSQL.
- Tech:** *Go, PostgreSQL, pgx/v5, Docker, golang-migrate, slog*

Full-Stack Projects

Narrative Guard: Brand Governance Platform

[Live](#) | [Source](#)

- Built a workspace-based brand governance platform with **role-based ownership enforcement** and **per-brand RAG namespaces** for workspace isolation.
- Built the audit flow end to end (score, verdict, findings, and an AI rewrite) in a responsive Next.js UI, with invite tokens stored as hashes and scheduled Convex actions handling async audit processing.
- Tech:** *Next.js, TypeScript, Convex, Convex Auth, Google Gemini, RAG, shadcn/ui, Vitest, React Testing Library*

Roadmap App

[Live](#) | [Source](#)

- Built a full-stack app with SSR, authentication, a REST API, and upvoting and commenting.
- Covered with tests using Jest and React Testing Library.
- Tech:** *Next.js, TypeScript, Drizzle ORM, PostgreSQL, Vercel, Jest, React Testing Library*

Technical Skills

Languages: TypeScript, JavaScript, Go, SQL

Frameworks: React.js, Next.js (SSR, ISR, SSG), Node.js, Express.js

Frontend: Tailwind CSS, shadcn/ui, Framer Motion

State Management: Context API, Zustand, React Query

Databases: PostgreSQL, MongoDB, MySQL, SQLite, Redis, Drizzle ORM

AI & Agents: Vercel AI SDK, Mastra AI, Google Gemini, RAG, Claude Code, Codex

DevOps/Infra: Docker, GitHub Actions, Vercel, Firebase, Appwrite, Convex, AWS (basic)

Testing & Tools: Git, Jest, Vitest, React Testing Library

Soft Skills: Communication, Ownership, Leadership, Collaboration, Analytical Thinking, Friendly & Approachable